



Greenlight Pro
Ballasted Solar Panel Racking System
Installation Manual



Green Roof Specialty Products and Sempergreen USA

Made by Contec

Introduction

Welcome to the Greenlight Pro Ballasted Solar Panel Racking System installation manual. This guide provides detailed instructions to ensure proper installation and operation of the Greenlight Pro system. By following these steps, you can ensure optimal performance and longevity.

Safety Notice

Before installation, ensure all personnel are trained and equipped with appropriate safety gear, including gloves, helmets, and steel-toed boots. Always adhere to local safety codes and regulations.

Pre-Installation Checklist

Before beginning installation, verify the following:

- All required system components are present.
- The installation site is clear of debris and structurally sound.
- Roof pitch and site conditions meet system specifications.
- Necessary tools and materials are on-site and functional.
- Safety measures, including fall protection, are in place.

Typical Installation rates

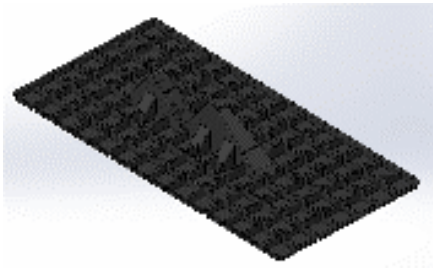
The following rates include the installation steps detailed in this manual, but do not include the loading, delivery, hoisting and staging of the materials, they should be allotted the appropriate additional time based on project conditions.

- **Over 100 Panels:** Base plates can be installed at a rate of 10 base plates per 2-person crew hour (5 base plates per man-hour).
- **Under 100 Panels:** Installation rates may be slower due to setup and logistical adjustments for smaller-scale projects.

Component Identification

Key components of the Greenlight Pro system include the following list. Be sure to verify quantities prior to mobilization:

- **Base Plate:** 78" x 38.6" x 2.4" (1980 mm x 980 mm x 60 mm).



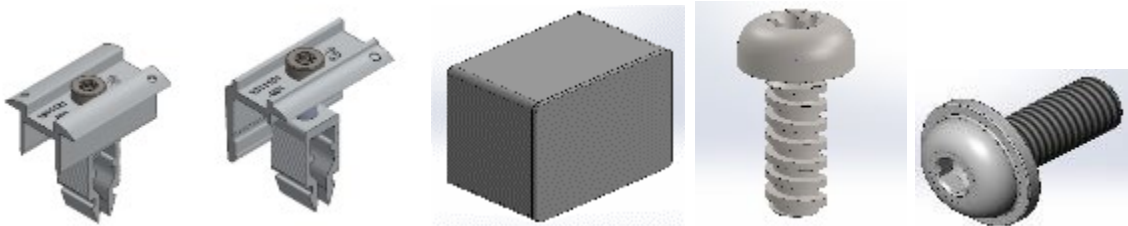
- **Knickfix Bracket:** Used to secure rails to base plates.



- **Rails:** Profile lengths of 6.6 ft to 11.3 ft (2.0 m to 3.4 m).



- **Small Components:** Screws, end caps and clamps for assembly.

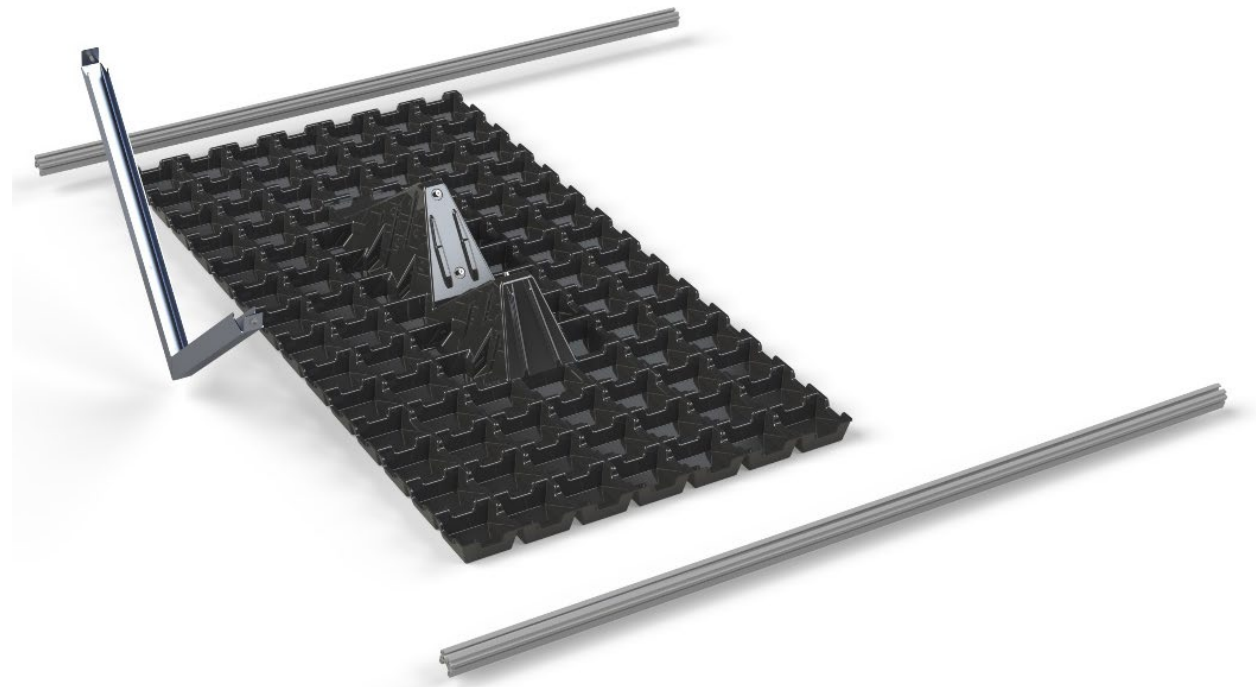


Installation Steps

1. Prepare the Site

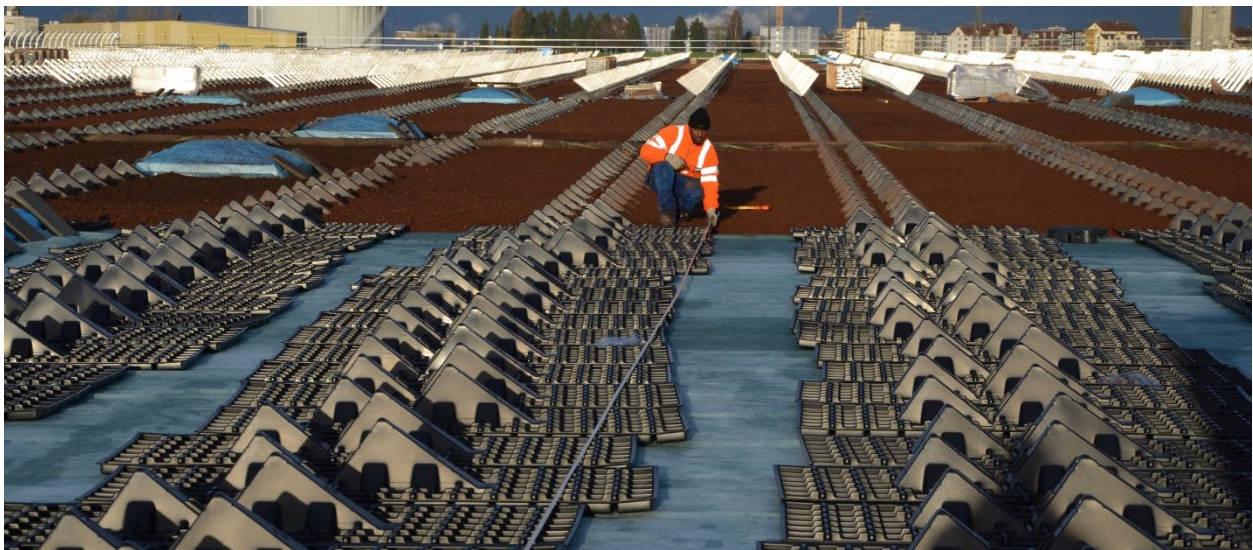
- Ensure the surface is level and stable.
- Confirm the layout with the shop drawings provided.
- **Protect the Waterproofing Membrane:** Place protective mats or other barriers as necessary to avoid damage to the waterproofing membrane during the installation process.

Baseplate, Knickfix v-bracket and two profile rails shown below:



2. Layout and Arrange Base Plates

- Use the shop drawings to determine the placement of the base plates.
- Arrange the base plates in the desired orientation:
 - **South Exposure:** For maximum sun exposure.
 - **East-West Orientation:** For balanced energy production throughout the day.
- Follow the proper spacing and overlap requirements per shop drawings.
- Use a rail and alignment guide to keep the plates straight during layout.
- **Protect the Membrane:** Handle all components carefully to avoid puncturing or damaging the underlying waterproofing layer.



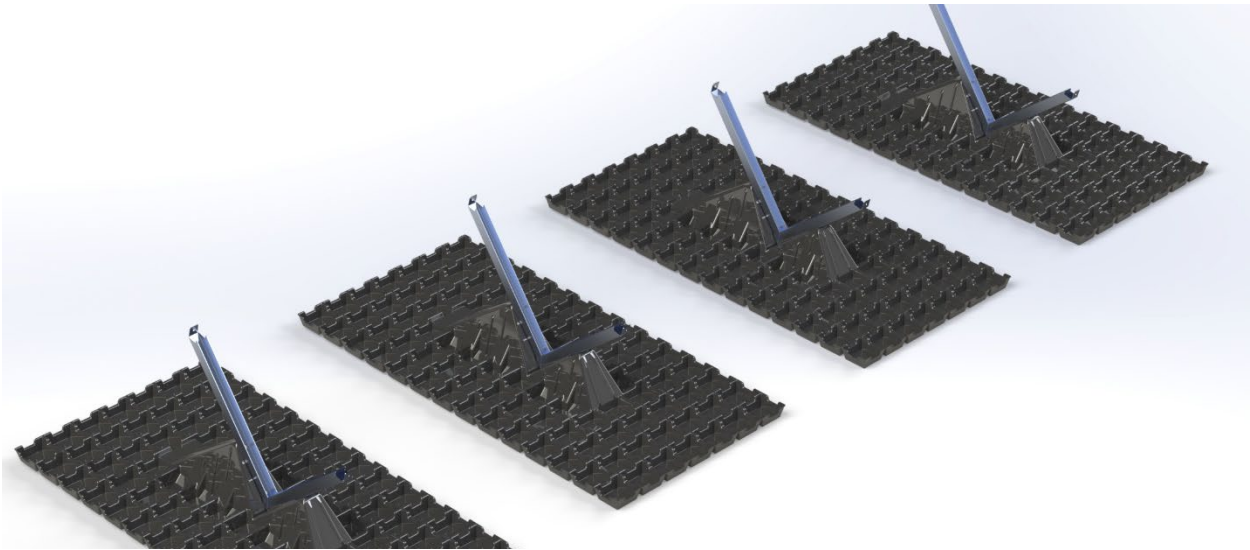
3. Ballast Base Plates

- Place ballast (e.g., sandbags or water bags (temporary) or growing media (permanent)) on the base plates to hold them in position.
- Verify alignment and adjust as needed.



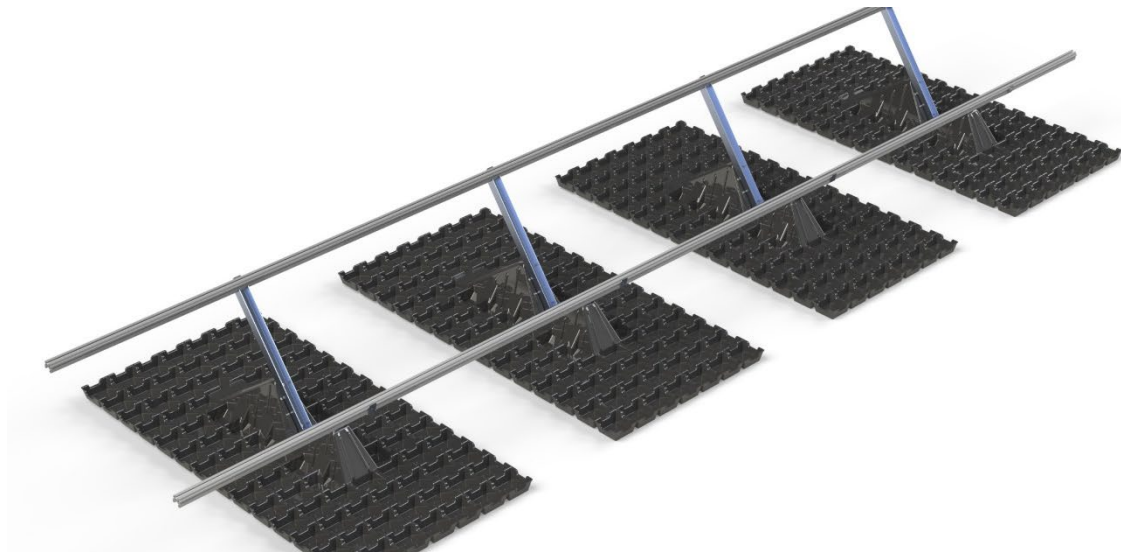
4. Install Knickfix v-rail Bracket

- Secure the Knickfix v-rail bracket to the base plates using the provided screws.
- Ensure each piece is properly arranged and aligned with fastening holes.
- Ensure each piece is properly tightly fastened to maintain stability.



5. Attach Rails

- Align the rails with the Knickfix v-rail bracket.
- Secure the rails to the hardware using screws.
- Check for proper alignment and levelness.
- After attaching the rails, ensure they are spaced correctly as per the shop drawings.



5. Attach caps, clamp and hardware

- Install all required end caps
- Install (or store for future) all middle clamps, end clamps, end caps and hardware for panel installation.

Shown below fully assembled with one (1) PV panel installed



Maintenance Guidelines

- Inspect the system annually for wear or damage.
 - Verify all screws, connectors, and clamps remain secure.
 - Clear debris from under and around panels to maintain drainage.
 - Clean PV panels regularly to maximize efficiency, following Manufacturer's instructions..
-

Troubleshooting Common Issues

- **Base Plate Misalignment:** Use a rail as a temporary guide to align baseplates and adjust spacing.
 - **Loose Connections:** Recheck screws and connectors for tightness.
 - **Water Drainage Issues:** Clear obstructions in drainage paths.
-

Post-Installation Checklist

- Confirm all components are securely fastened.
 - Verify alignment matches the shop drawings.
 - Ensure temporary ballast is replaced with permanent growing media or approved ballasting material at appropriate depths.
-

Post-Installation Cleanup

- **Metal Debris and Fasteners:** Collect and remove all metal debris, screws, and fasteners from the installation area to prevent damage to the waterproofing membrane.
 - **Inspect** the work area thoroughly to ensure no leftover materials or tools are left on the roof.
 - Properly **haul and dispose** of debris off-site according to local regulations.
-

Technical Data: Greenlight Pro (4 mm thickness)

- **Base Plate Dimensions:** 78" x 38.6" x 2.4" (1980 mm x 980 mm x 60 mm).
 - **Weight per Unit:**
 - Without ballast, without PV module: 26.5 lbs. (12 kg).
 - Equivalent weight: 1.2 lbs./sq ft (6 kg/m²).
 - **Water Storage Volume:** 10.5 gallons (39.5 liters), unfilled.
 - **Compressive Strength:**
 - Unfilled: 5,620 lbf (25 kN/m²).
 - Backfilled (flush): 15,736 lbf (70 kN/m²).
 - **Water Drainage Capacity:**
 - $i = 0.01$ (1% gradient): 0.4 l/(m*s).
 - $i = 0.02$ (2% gradient): 0.6 l/(m*s).
 - $i = 0.05$ (5% gradient): 1.0 l/(m*s).
 - **Material:** HDPE, Magnelis, aluminum, stainless steel.
 - **Standard Module Inclination:** 10°, 15°, or 20° (optional 5° to max. 35° on request).
 - **Roof Connection:** No structural roof connection required.
 - **Ballasting:** Calculated by Contec AG based on wind zone plan.
 - **Roof Pitch:**
 - Approved up to 5°.
 - Over 5° requires technical approval by Contec AG.
 - **System Components:**
 - 1x 4mm Greenlight pro base plate.
 - 1x Knickfix v-rail.
 - 6.6 ft to 11.3 ft (2.0 m to 3.4 m profile rail).
 - Small material (screws, profile connectors, clamps).
 - **Warranty:** 10-year system warranty on substructure ex-works (valid from delivery date).
-